

UNIVERSITY OF OSLO

CTaps

And gaps when supporting QUIC through
TAPS

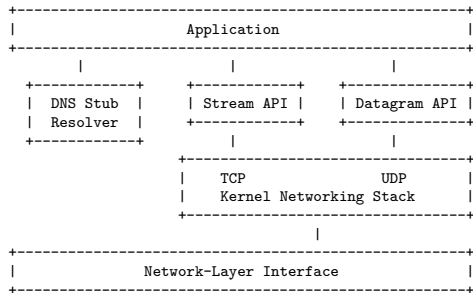
Ingebrigt Hovind

Department of Informatics

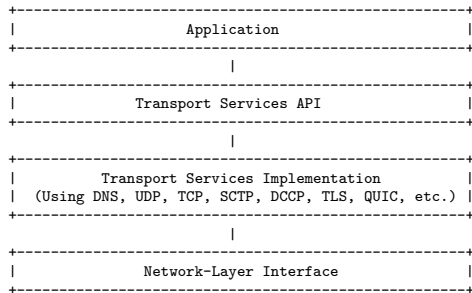
July 22nd, 2026



What is Transport Services?



(a) Socket Programming



(b) TAPS

Source: RFC 9621

Vision of TAPS

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 - Common policies, utilizing underlying mechanisms to achieve these
 - E.g. stream priority, security preferences, DSCP

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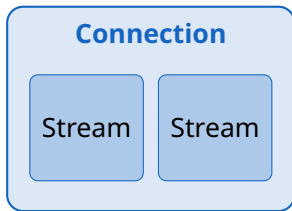
- A single interface, for more than just transport layer protocols:
 - Common policies, utilizing underlying mechanisms to achieve these
 - E.g. stream priority, security preferences, DSCP
- Easier protocol adoption due to Happy Eyeballs-style racing

What is CTaps?

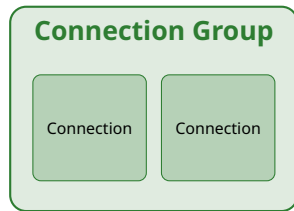
What is CTaps?

- C implementation of TAPS
- QUIC support
 - With certain gaps

Stream closing semantics



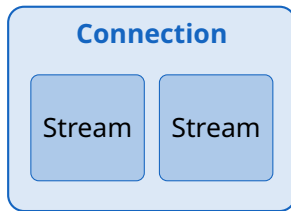
(a) QUIC



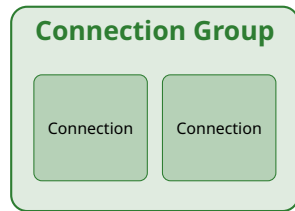
(b) TAPS

Stream closing semantics

- QUIC Connections map to Connection Groups



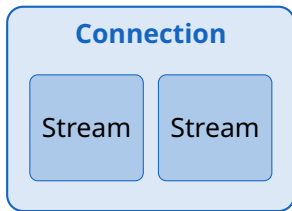
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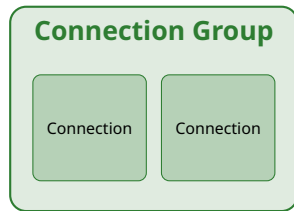
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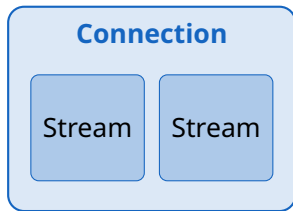
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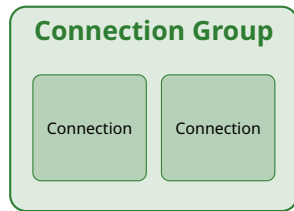
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- No equivalent mapping for TCP



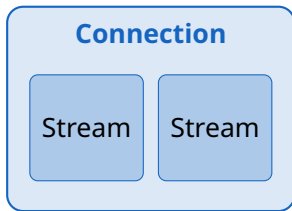
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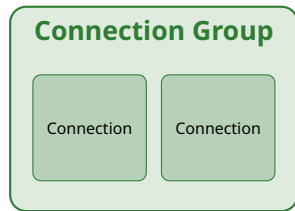
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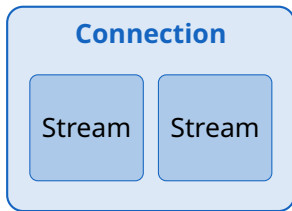
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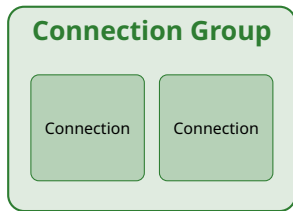
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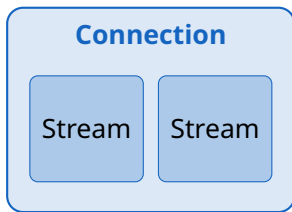
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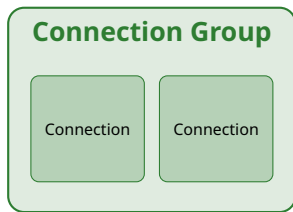
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Stream closing semantics

- QUIC Connections map to Connection Groups
- Streams map to Connections
- No equivalent mapping for TCP
- On `close()`:
 - TCP wants to initiate graceful teardown
 - QUIC wants to send FIN but remain open



(a) QUIC



(b) TAPS

Losing granularity

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- Aborting streams:
 - QUIC:
 - ◇ STOP_SENDING
 - ◇ RESET_STREAM
 - TAPS:
 - ◇ Just a single Abort()
- Directional stream establishment:
 - TAPS can only set directionality on initial establishment
 - Cloning and new streams happens later

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- **Closing semantics**

- Clearly define expected behavior
- Closing Connection Group achieves same behavior accross protocols
 - ◇ Client applications utilizing TAPS must follow this advice

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- Some gaps in QUIC support do exist
 - But can be mitigated with careful design
 - Relevant for later projects
 - ◇ TAPS-based networking in C++ standard library?

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- Application level protocols can run over QUIC with built-in fallback
- Some gaps in QUIC support do exist
 - But can be mitigated with careful design
 - Relevant for later projects
 - ◇ TAPS-based networking in C++ standard library?
- Both the source code and thesis text can be found on Github:
 - <https://github.com/ikhovind/CTaps>
 - <https://github.com/ikhovind/master-thesis/releases>